

Gimbal Kinematics of a Line Segment on a Rotating Rectangle

Two-Axis Pivot Rotation & Homogeneous Transformation

Classification: Internal

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July 7, 2026

Objective: Extend the single-axis segment rotation to a two-axis gimbal: the segment rotates by ϕ about its first gimbal axis $\mathbf{u}_1$, then by ω about a perpendicular second axis $\mathbf{u}_2$ ($\omega \in (0, \pi/2)$), both passing through the top endpoint. Derive the parametric trajectory $C(\theta, \phi, \omega)$ and the 4×4 homogeneous transformation matrix from the world frame to a body-fixed frame.

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1 Problem Setup

A rectangle rotates about its left vertical edge, which coincides with the z -axis. A line segment (perpendicular to the rectangle plane, i.e. parallel to the z -axis) is attached to the rectangle. The segment undergoes a two-axis gimbal rotation about axes passing through its top endpoint.

- The rectangle's rotation edge is the z -axis $(0, 0, z)$.
- At $\theta = 0$ the rectangle lies in the xz -plane; the segment center is at $(r, 0, h)$, at distance r from the z -axis.
- h is the height of the center point, L is the half-length (full length $2L$).
- **First gimbal axis** (in world coordinates, rotated with the rectangle): $\mathbf{u}_1 = (\cos \theta, \sin \theta, 0)$. The segment rotates by ϕ about this axis through its top endpoint.
- **Second gimbal axis** (perpendicular to the first, also fixed in world space once θ is chosen): $\mathbf{u}_2 = (-\sin \theta, \cos \theta, 0)$. The segment rotates by ω about this axis through its top endpoint, with $\omega \in (0, \frac{\pi}{2})$.
- $\mathbf{u}_1 \cdot \mathbf{u}_2 = 0$; both are horizontal unit vectors in the xy -plane. $\mathbf{u}_2$ is $\mathbf{u}_1$ rotated by $+90^\circ$ about z .

We seek the parametric equation of the *center point* of the segment after all three rotations (θ, ϕ, ω) , the target-frame axes, and the homogeneous transformation matrix.

2 Reference Positions

2.1 Center point after rectangle rotation only

$$P(\theta) = (r \cos \theta, r \sin \theta, h).$$

2.2 Top endpoint (gimbal centre)

Since the segment is initially vertical, the top endpoint is $+L$ above the center:

$$T(\theta) = P(\theta) + (0, 0, L) = (r \cos \theta, r \sin \theta, h + L). \quad (1)$$

2.3 Vector from top to center

$$\mathbf{d} = P - T = (0, 0, -L). \quad (2)$$

3 Rotation Matrices via Rodrigues' Formula

Rodrigues' formula for a rotation by α about a unit axis $\mathbf{u}$:

$$\mathbf{R}_{\mathbf{u}}(\alpha) = I + \sin \alpha [\mathbf{u}]_{\times} + (1 - \cos \alpha) [\mathbf{u}]_{\times}^2, \quad (3)$$

where $[\mathbf{u}]_{\times}$ is the skew-symmetric cross-product matrix.

3.1 First rotation — $\mathbf{R}_1 = \mathbf{R}_{\mathbf{u}_1}(\phi)$

Let $s_\theta = \sin \theta$, $c_\theta = \cos \theta$, $s_\phi = \sin \phi$, $c_\phi = \cos \phi$, $s_\omega = \sin \omega$, $c_\omega = \cos \omega$.

$$\mathbf{R}_1 = \begin{bmatrix} 1 - (1 - c_\phi)s_\theta^2 & (1 - c_\phi)s_\theta c_\theta & s_\phi s_\theta \\ (1 - c_\phi)s_\theta c_\theta & 1 - (1 - c_\phi)c_\theta^2 & -s_\phi c_\theta \\ -s_\phi s_\theta & s_\phi c_\theta & c_\phi \end{bmatrix}. \quad (4)$$

3.2 Second rotation — $\mathbf{R}_2 = \mathbf{R}_{\mathbf{u}_2}(\omega)$

$$\mathbf{R}_2 = \begin{bmatrix} 1 - (1 - c_\omega)c_\theta^2 & -(1 - c_\omega)s_\theta c_\theta & s_\omega c_\theta \\ -(1 - c_\omega)s_\theta c_\theta & 1 - (1 - c_\omega)s_\theta^2 & s_\omega s_\theta \\ -s_\omega c_\theta & -s_\omega s_\theta & c_\omega \end{bmatrix}. \quad (5)$$

Constraint: $\omega \in (0, \pi/2)$. The second gimbal angle is restricted to the first quadrant ($\omega \in (0^\circ, 90^\circ)$), modelling a physical hinge that cannot close past zero or open beyond horizontal.

3.3 Composite rotation — $\mathbf{R} = \mathbf{R}_2 \mathbf{R}_1$

Applying ϕ then ω yields the combined orientation of the segment body. The third column (segment direction $\hat{\mathbf{x}}_T$) is particularly simple:

$$\hat{\mathbf{x}}_T = \mathbf{R} \mathbf{e}_z = \begin{pmatrix} s_\theta s_\phi + c_\theta c_\phi s_\omega \\ -c_\theta s_\phi + s_\theta c_\phi s_\omega \\ c_\phi c_\omega \end{pmatrix}. \quad (6)$$

The full 3×3 product $\mathbf{R}_2 \mathbf{R}_1$ is unwieldy in closed form; compose $\mathbf{R}_1$ and $\mathbf{R}_2$ numerically in implementation.

4 Trajectory of the Segment Centre

The centre in body-frame coordinates is $(0, 0, -L)$. After the composite rotation and translation by T :

$$\begin{aligned} C(\theta, \phi, \omega) &= T + \mathbf{R} (0, 0, -L)^\top = T - L \hat{\mathbf{x}}_T \\ &= \begin{pmatrix} r \cos \theta - L \sin \theta \sin \phi - L \cos \theta \cos \phi \sin \omega \\ r \sin \theta + L \cos \theta \sin \phi - L \sin \theta \cos \phi \sin \omega \\ h + L (1 - \cos \phi \cos \omega) \end{pmatrix}. \end{aligned} \quad (7)$$

$$C(\theta, \phi, \omega) = \begin{pmatrix} r \cos \theta - L \sin \theta \sin \phi - L \cos \theta \cos \phi \sin \omega \\ r \sin \theta + L \cos \theta \sin \phi - L \sin \theta \cos \phi \sin \omega \\ h + L (1 - \cos \phi \cos \omega) \end{pmatrix}. \quad (8)$$

Table 1: Parameter definitions

Symbol	Meaning	Range
θ	Rectangle rotation about z -axis	$[0, 2\pi)$
ϕ	First gimbal rotation about $\mathbf{u}_1$ (through T)	$[0, 2\pi)$
ω	Second gimbal rotation about $\mathbf{u}_2$ (through T)	$(0, \pi/2)$
r	Distance from segment to z -axis	
h	Height of segment center (at $\theta = \phi = \omega = 0$)	
L	Half-length of the segment	

5 Verification

Case 1: $\phi = \omega = 0$ (no gimbal rotation)

$$C(\theta, 0, 0) = (r \cos \theta, r \sin \theta, h) = P(\theta).$$

The center returns to its position on the rotating rectangle.

Case 2: $\omega = 0$ (single-axis rotation about $\mathbf{u}_1$)

$$C(\theta, \phi, 0) = \begin{pmatrix} r \cos \theta - L \sin \theta \sin \phi \\ r \sin \theta + L \cos \theta \sin \phi \\ h + L(1 - \cos \phi) \end{pmatrix},$$

which matches the single-axis result of the companion document.

Case 3: $\phi = 0$ (single-axis rotation about $\mathbf{u}_2$)

$$C(\theta, 0, \omega) = \begin{pmatrix} r \cos \theta - L \cos \theta \sin \omega \\ r \sin \theta - L \sin \theta \sin \omega \\ h + L(1 - \cos \omega) \end{pmatrix}.$$

The ω rotation tilts the segment in the direction of $-\mathbf{u}_1$ (since $\mathbf{u}_2$ is at $+90^\circ$ from $\mathbf{u}_1$). For $\theta = 0$ this gives $C(0, 0, \omega) = (r - L \sin \omega, 0, h + L(1 - \cos \omega))$ — the center traces a circle of radius L in the xz -plane.

Case 4: $\phi = \pi/2$, ω arbitrary (segment tilted then hinged)

$$C(\theta, \frac{\pi}{2}, \omega) = \begin{pmatrix} r \cos \theta - L \sin \theta \\ r \sin \theta + L \cos \theta \\ h + L \end{pmatrix}.$$

The z -coordinate becomes $h + L$ (segment horizontal); ω has no further effect because the segment is already perpendicular to $\mathbf{u}_2$'s plane of action.

Case 5: $\omega = \pi/2$ (second hinge fully open)

$$C(\theta, \phi, \frac{\pi}{2}) = \begin{pmatrix} r \cos \theta - L \sin \theta \sin \phi - L \cos \theta \cos \phi \\ r \sin \theta + L \cos \theta \sin \phi - L \sin \theta \cos \phi \\ h + L \end{pmatrix}.$$

At $\omega = \pi/2$ the segment lies flat in the xy -plane (height $h + L$). The x, y components simplify to a rotation of $(-L, 0, 0)$ by θ about z plus the ϕ -tilted top endpoint: $C = (r \cos \theta, r \sin \theta, h + L) + \mathbf{R}_1|_{z=0} (-L, 0, 0)^\top$.

Case 6: Distance preservation

For fixed (θ, ϕ, ω) the distance from C to the top endpoint T is always L : $\|C - T\| = \|L \hat{\mathbf{x}}_T\| = L$. The center moves on a sphere of radius L centered at $T(\theta)$.

6 Interpretation

For fixed θ , $(\phi, \omega) \mapsto C$ is a composition of two rotations of $(0, 0, -L)$ about orthogonal axes, yielding a point on the sphere of radius L centered at $T(\theta)$. The z -coordinate $h + L(1 - c_\phi c_\omega)$ ranges from h (at $\phi = \pm\pi/2$ or $\omega = \pi/2$) to $h + 2L$ (at $\phi = \omega = \pi$, though ω is bounded below $\pi/2$ so the maximum is at $\phi = \pi, \omega \rightarrow 0$).

For fixed ϕ and ω , varying θ sweeps a curve on a cylindrical surface of approximate radius r around the z -axis, with a compound sinusoidal ripple of amplitude $L(\sin \phi + \cos \phi \sin \omega)$ from both gimbal angles.

7 Target Frame

7.1 Frame definition

Attach a right-handed coordinate frame $\{T\}$ to the segment:

- **Origin:** the segment center point $C(\theta, \phi, \omega)$.
- $\hat{\mathbf{x}}_T$: along the segment, bottom to top. This is $\mathbf{R} \mathbf{e}_z$ — the body z -axis after both rotations.
- $\hat{\mathbf{y}}_T$: the image of $\mathbf{u}_1$ after the second rotation. Since the first ϕ -rotation carries the body and the second ω -rotation tilts it about $\mathbf{u}_2$, the $\hat{\mathbf{y}}_T$ axis is $\mathbf{R}_2 \mathbf{u}_1 = \mathbf{R}_2(\omega) \mathbf{u}_1$.
- $\hat{\mathbf{z}}_T$: $\hat{\mathbf{x}}_T \times \hat{\mathbf{y}}_T$ (completing the right-handed system).

7.2 Axes in world coordinates

$$\hat{\mathbf{x}}_T = \begin{pmatrix} \sin \theta \sin \phi + \cos \theta \cos \phi \sin \omega \\ -\cos \theta \sin \phi + \sin \theta \cos \phi \sin \omega \\ \cos \phi \cos \omega \end{pmatrix} \quad (9)$$

$$\hat{\mathbf{y}}_T = \mathbf{R}_2(\omega) \mathbf{u}_1 = \begin{pmatrix} \cos \theta \cos \omega \\ \sin \theta \cos \omega \\ -\sin \omega \end{pmatrix} \quad (10)$$

$$\hat{\mathbf{z}}_T = \hat{\mathbf{x}}_T \times \hat{\mathbf{y}}_T = \begin{pmatrix} -\cos \theta \sin \phi \sin \omega - \sin \theta \cos \phi \cos^2 \omega - \sin \theta \cos \phi \sin^2 \omega \\ \cos \theta \cos \phi \cos^2 \omega + \cos \theta \cos \phi \sin^2 \omega - \sin \theta \sin \phi \sin \omega \\ \cos \theta \sin \theta \sin \phi \cos \omega - \sin \theta \cos \theta \sin \phi \cos \omega + \sin \phi \cos \omega \end{pmatrix}. \quad (11)$$

Simplifications: $\cos^2 \omega + \sin^2 \omega = 1$, and $\cos \theta \sin \theta \sin \phi \cos \omega - \sin \theta \cos \theta \sin \phi \cos \omega = 0$. Hence

$$\hat{\mathbf{z}}_T = \begin{pmatrix} -\cos \theta \sin \phi \sin \omega - \sin \theta \cos \phi \\ \cos \theta \cos \phi - \sin \theta \sin \phi \sin \omega \\ \sin \phi \cos \omega \end{pmatrix}. \quad (12)$$

Verification. $\|\hat{\mathbf{y}}_T\| = \cos^2 \theta \cos^2 \omega + \sin^2 \theta \cos^2 \omega + \sin^2 \omega = \cos^2 \omega + \sin^2 \omega = 1$. $\hat{\mathbf{x}}_T \cdot \hat{\mathbf{y}}_T = 0$: the cross terms cancel, leaving $c_\phi s_\omega c_\omega - c_\phi s_\omega c_\omega = 0$.

7.3 Rotation matrix

Assemble these axes as columns (maps target $\rightarrow$ world):

$$\mathbf{R}_T = [\hat{\mathbf{x}}_T \quad \hat{\mathbf{y}}_T \quad \hat{\mathbf{z}}_T]. \quad (13)$$

For a world-frame vector $\mathbf{w}$, its target-frame representation is $\mathbf{w}_T = \mathbf{R}_T^\top \mathbf{w}$, since $\mathbf{R}_T$ is orthogonal.

8 Homogeneous Transformation Matrix

8.1 Translation component

A point $\mathbf{p}$ in world coordinates maps to target coordinates as:

$$\mathbf{p}_T = \mathbf{R}_T^\top (\mathbf{p} - C(\theta, \phi, \omega)).$$

Compute $\mathbf{R}_T^\top C$ using the closed-form axes and C :

$$\begin{aligned} \mathbf{R}_T^\top C &= \begin{bmatrix} \hat{\mathbf{x}}_T^\top \\ \hat{\mathbf{y}}_T^\top \\ \hat{\mathbf{z}}_T^\top \end{bmatrix} \begin{pmatrix} r \cos \theta - L \sin \theta s_\phi - L \cos \theta c_\phi s_\omega \\ r \sin \theta + L \cos \theta s_\phi - L \sin \theta c_\phi s_\omega \\ h + L(1 - c_\phi c_\omega) \end{pmatrix} \\ &= \begin{pmatrix} -h c_\phi c_\omega + L(1 - c_\phi c_\omega) \\ r \\ h s_\phi c_\omega - L c_\phi s_\omega \end{pmatrix}. \end{aligned} \quad (14)$$

The θ -dependence cancels entirely in the translation component (as expected: the top-endpoint pivot is blind to the rectangle's global rotation).

8.2 Full 4×4 matrix

The homogeneous transformation ${}^{\text{world}}\mathbf{T}_{\text{target}}$ maps world to target coordinates:

$${}^{\text{world}}\mathbf{T}_{\text{target}} = \begin{bmatrix} \mathbf{R}_T^\top & -\mathbf{R}_T^\top C \\ \mathbf{0}_{1 \times 3} & 1 \end{bmatrix}. \quad (15)$$

For a world point $\mathbf{p}_{\text{world}}$:

$$\begin{pmatrix} \mathbf{p}_{\text{target}} \\ 1 \end{pmatrix} = {}^{\text{world}}\mathbf{T}_{\text{target}} \begin{pmatrix} \mathbf{p}_{\text{world}} \\ 1 \end{pmatrix}.$$

8.3 Verification

- The origin C maps to $(0, 0, 0)$: $\mathbf{R}_T^T(C - C) = \mathbf{0}$.
- Bottom endpoint $B = C - L \hat{\mathbf{x}}_T$ maps to $(-L, 0, 0)$.
- Top endpoint $T = C + L \hat{\mathbf{x}}_T$ maps to $(L, 0, 0)$.
- Point $C + \hat{\mathbf{y}}_T$ maps to $(0, 1, 0)$.
- At $\phi = \omega = 0$: $\mathbf{R}_T = [\mathbf{e}_z \quad \mathbf{u}_1 \quad \mathbf{e}_z \times \mathbf{u}_1]$ with origin at $P(\theta)$; the z -axis is aligned with the segment.
- At $\omega = 0$, the results reduce to the single-axis case of the companion document.

9 Summary

The segment center trajectory $C(\theta, \phi, \omega)$ is given by (8). The homogeneous transformation ${}^{\text{world}}\mathbf{T}_{\text{target}}$ is built from axes (9)–(12) and the translation (14).

For implementation:

1. Compute T from (1).
2. Build $\mathbf{R}_1$ and $\mathbf{R}_2$ from (4) and (5).
3. Numerically compose $\mathbf{R} = \mathbf{R}_2\mathbf{R}_1$.
4. The centre position is $T - L\mathbf{R}\mathbf{e}_z$; use the closed-form (8) to avoid round-off.
5. Build $\mathbf{R}_T$ from axes (9)–(12); the translation column is $-\mathbf{R}_T^T C$ from (14).
6. Compose the 4×4 homogeneous matrix (15).

10 Sequential Motion Example

10.1 Motion model

Consider a sequential task where the three degrees of freedom are actuated one at a time, from rest to the final pose. With parameters $r = 2$, $L = 1.5$, $h = 0$, the motion proceeds through three stages:

Stage	θ	ϕ	ω	Description
1	0	$0 \rightarrow \pi/4$	0	First gimbal opens
2	0	$\pi/4$	$0 \rightarrow \pi/2$	Second gimbal opens
3	$0 \rightarrow 2\pi/3$	$\pi/4$	$\pi/2$	Rectangle drives to final pose

The constraint $\theta \in (0, 2\pi/3)$ and $\omega \in (0, \pi/2)$ are respected at all times.

10.2 Visual overview

Figure 1 shows the four key poses: start, end of each stage. The target frame $\{\hat{\mathbf{x}}_T, \hat{\mathbf{y}}_T, \hat{\mathbf{z}}_T\}$ is drawn at the segment center.

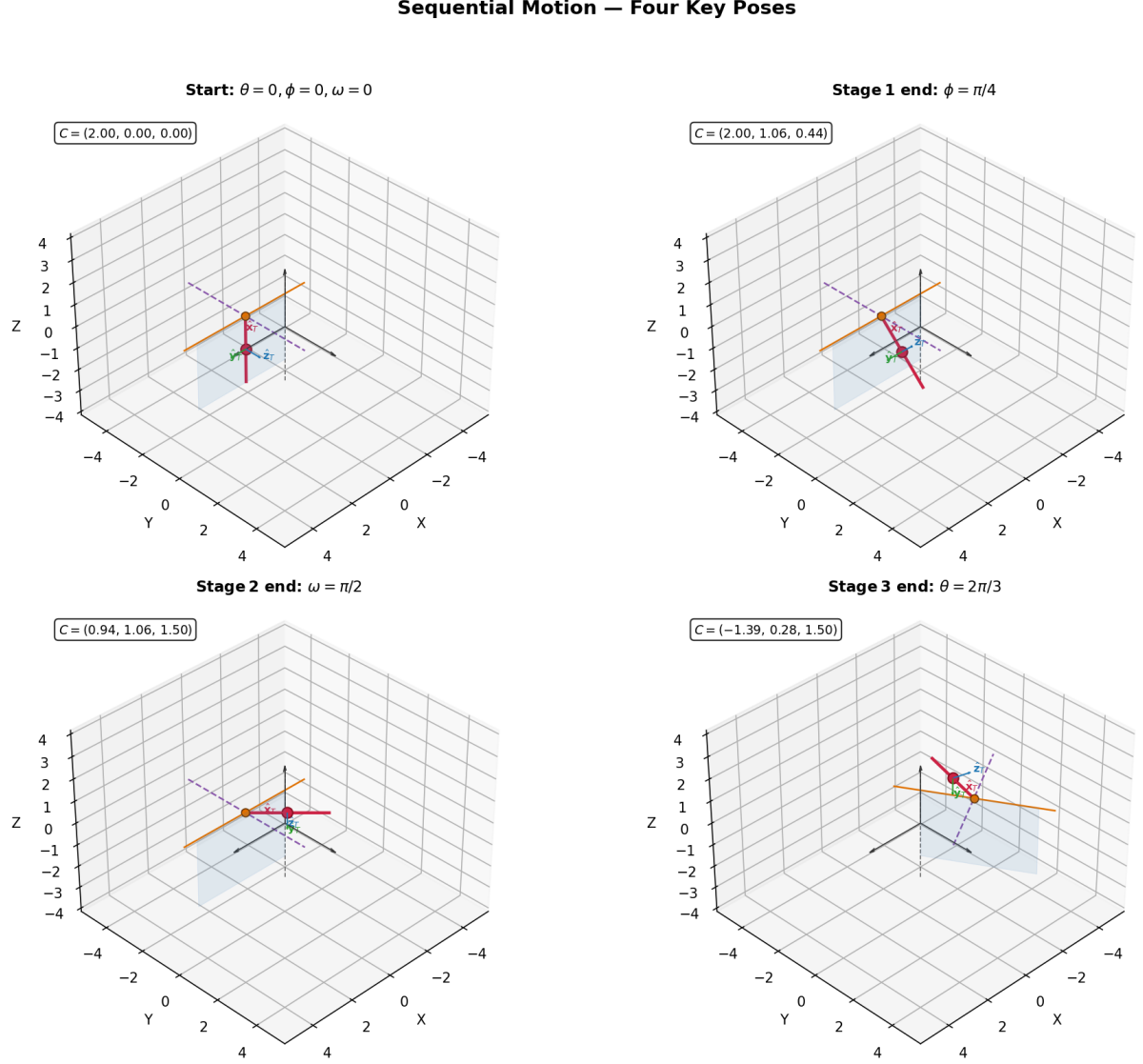


Figure 1: Four key poses during the sequential motion. Top-left: initial configuration ($\theta=0, \phi=0, \omega=0$). Top-right: after Stage 1 ($\phi=\pi/4$). Bottom-left: after Stage 2 ($\omega=\pi/2$). Bottom-right: final pose ($\theta=2\pi/3, \phi=\pi/4, \omega=\pi/2$).

10.3 Center point trajectory

Figure 2 tracks the center point C through the full motion. The three stages are color-coded; the blue curve (Stage 3) dominates the lateral displacement as the rectangle sweeps through θ .

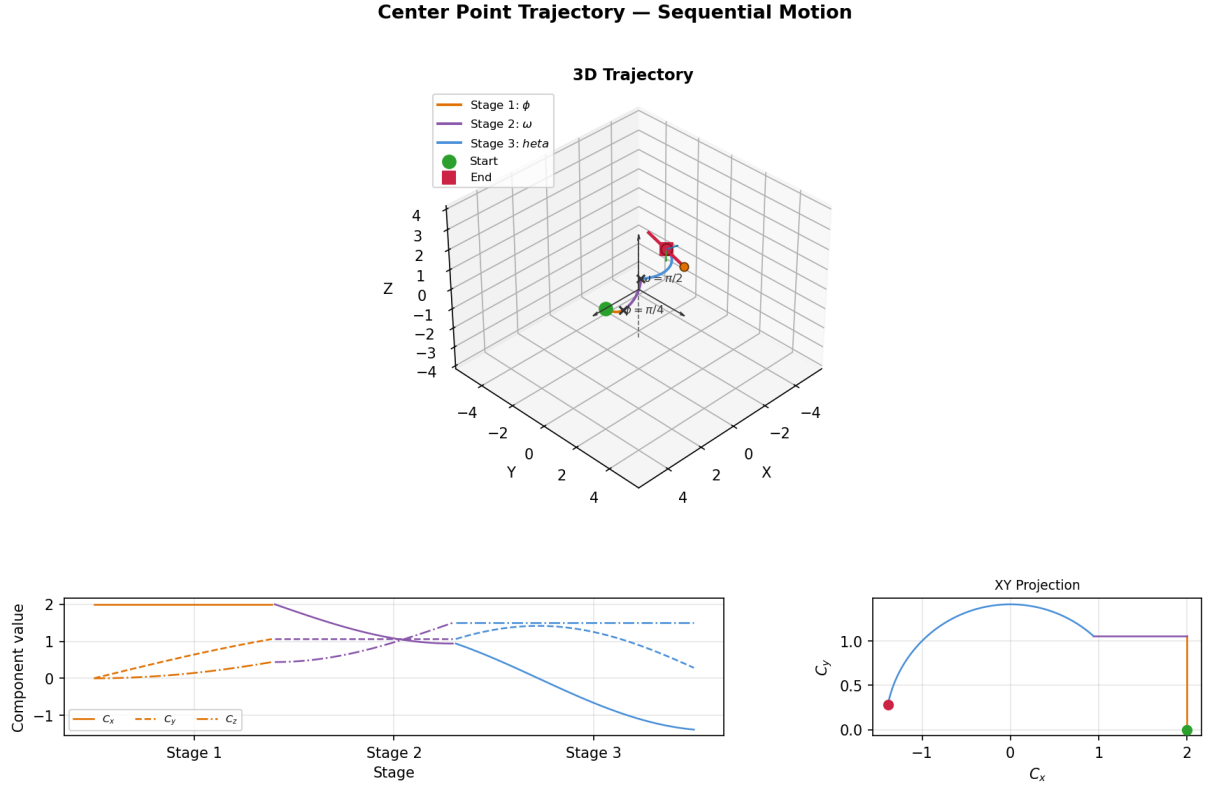


Figure 2: Center point trajectory. Left: 3D view. Right: component-wise evolution across stages and XY projection.

The stage-end coordinates are:

$$\text{Stage 1 end } (\theta=0, \phi=\frac{\pi}{4}, \omega=0) : C = (r, r + L\frac{\sqrt{2}}{2}, h + L(1 - \frac{\sqrt{2}}{2}))$$

$$\text{Stage 2 end } (\theta=0, \phi=\frac{\pi}{4}, \omega=\frac{\pi}{2}) : C = (r - L\frac{\sqrt{2}}{2}, r + L\frac{\sqrt{2}}{2}, h + L)$$

$$\text{Stage 3 end } (\theta=\frac{2\pi}{3}, \phi=\frac{\pi}{4}, \omega=\frac{\pi}{2}) : C_f = \begin{pmatrix} -\frac{r}{2} + L\frac{\sqrt{2} - \sqrt{6}}{4} \\ \frac{r\sqrt{3}}{2} - L\frac{\sqrt{2} + \sqrt{6}}{4} \\ h + L \end{pmatrix}$$

10.4 Time-varying rigid-body transform

The homogeneous transformation from world to target coordinates is ${}^{\text{world}}\mathbf{T}_{\text{target}} = [\mathbf{R}_T^T \ -\mathbf{R}_T^T C]$ with $\mathbf{R}_T^T$ from (9)–(12) and $-\mathbf{R}_T^T C$ from (14).

Substituting the stage-specific parameter trajectories yields $T(t)$ for each stage. Let $S_t = \sin t$, $C_t = \cos t$, and $\alpha = \sqrt{2}/2$.

10.4.1 Stage 1 — $\theta=0, \phi=t, \omega=0$

The first gimbal opens while the rectangle and second hinge are static. The transform reduces to the single-axis case:

$$T_1(t) = \left[\begin{array}{ccc|c} 0 & -S_t & C_t & L(1 - C_t) - h C_t \\ 1 & 0 & 0 & -r \\ 0 & C_t & S_t & -(L + h)S_t \\ \hline 0 & 0 & 0 & 1 \end{array} \right], \quad t \in [0, \frac{\pi}{4}]. \quad (16)$$

10.4.2 Stage 2 — $\theta=0$, $\phi=\pi/4$, $\omega=t$

The second gimbal opens; the first is held at $\pi/4$:

$$T_2(t) = \left[\begin{array}{ccc|c} \alpha S_t & -\alpha & \alpha C_t & L(1 - \alpha C_t) - h \alpha C_t \\ C_t & 0 & -S_t & -r \\ -\alpha S_t & \alpha & \alpha C_t & h \alpha S_t \\ \hline 0 & 0 & 0 & 1 \end{array} \right], \quad t \in [0, \frac{\pi}{2}]. \quad (17)$$

10.4.3 Stage 3 — $\theta=t$, $\phi=\pi/4$, $\omega=\pi/2$

The rectangle drives to its final position; both gimbals are at their limit:

$$T_3(t) = \left[\begin{array}{ccc|c} \alpha(S_t+C_t) & \alpha(S_t-C_t) & 0 & L - h \alpha C_t \\ 0 & 0 & -1 & -r \\ \alpha(C_t-S_t) & \alpha(S_t+C_t) & 0 & h \alpha S_t \\ \hline 0 & 0 & 0 & 1 \end{array} \right], \quad t \in [0, \frac{2\pi}{3}]. \quad (18)$$

Figure 3 visualizes the twelve non-trivial entries of $T(t)$ as functions of the motion progress parameter.

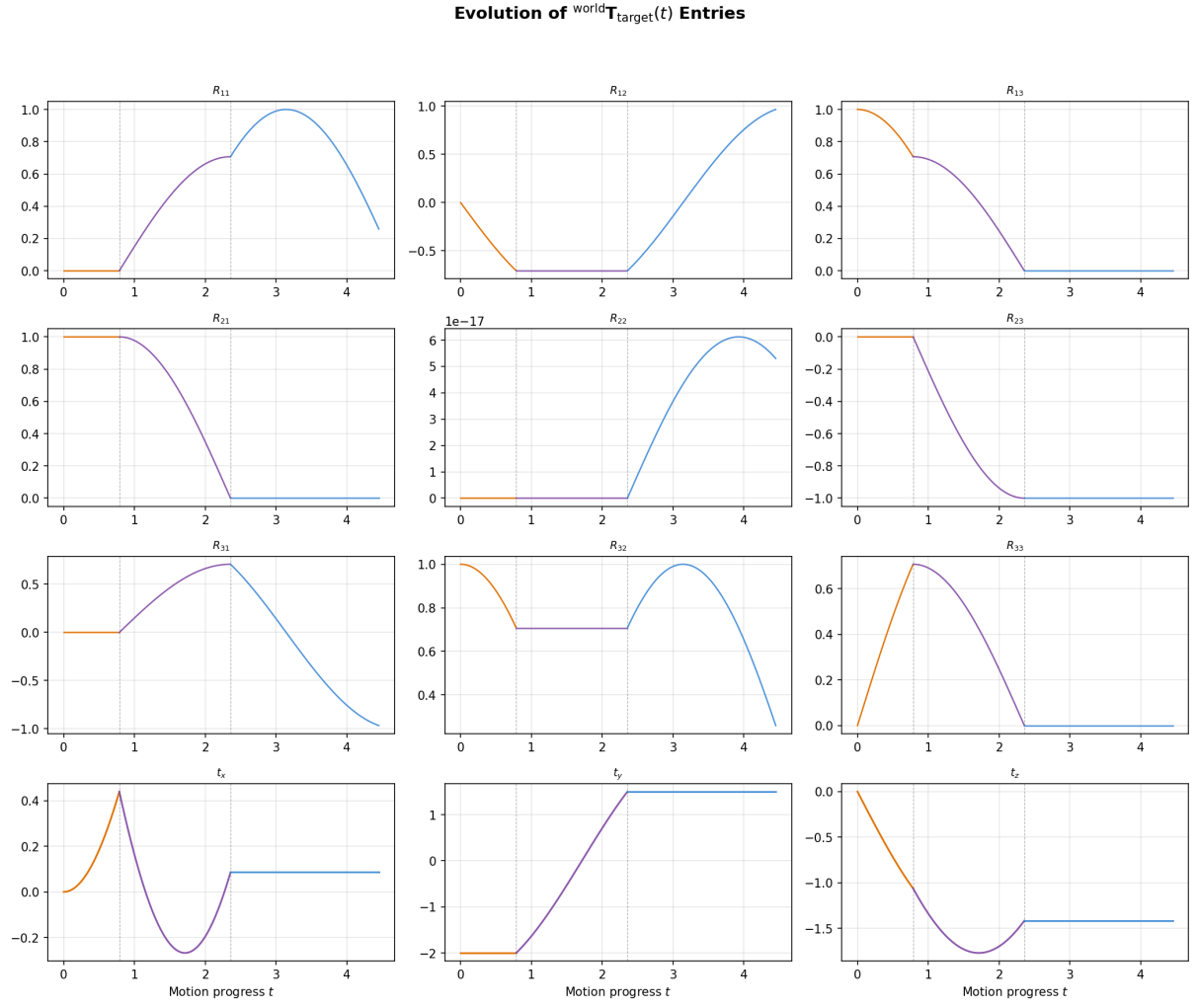


Figure 3: Evolution of ${}^{\text{world}}\mathbf{T}_{\text{target}}(t)$ entries through the three-stage sequential motion. Top row: rotation block entries R_{ij}^T . Bottom row: translation entries t_x, t_y, t_z . Vertical dashed lines mark stage boundaries. Parameters: $r=2, L=1.5, h=0$.